

Homework 1 — what counts

CIVL 6962 · Machine Learning in Transportation · due 09/18/2026 · the requirements are on the Class 3 slide "Now build one"; this sheet is what they mean

Do this today, before anything else. Deploy an empty app — one line, `st.title("HW1")` — and get the URL working. Then go back and build the dashboard. First-time deployment fails for reasons that have nothing to do with your code (requirements, repo permissions, file paths, file size), and those are cheap to fix on day 1 and expensive on the last night. **A dashboard that runs on your laptop and not on the URL scores as a dashboard that does not run.**

WHAT COUNTS AS ONE OF THE THREE CONTROLS

Change the widget. Do the NUMBERS on the chart change?
If yes, it is a control. If only the appearance changed, it is not.

WIDGET	WHY
YES date-range slider	different rows go into the chart
YES sensor / segment multiselect	different series are computed
YES metric radio — speed / flow / occupancy	a different column is plotted
YES aggregation selectbox — 5-min / hourly / daily	the data is re-aggregated
NO colour theme, chart height, "show legend"	same numbers, different paint
NO three widgets that all restyle the same one chart	this is exactly what the slide is warning about

WHAT COUNTS AS "TWO DIFFERENT KINDS" OF CHART

Line, bar, scatter, histogram, box plot, heatmap, map — these are different kinds. **Two lines drawn on one axis is one kind, not two.** A table (`st.dataframe`) is not a chart; it can be your fourth element but it does not satisfy the requirement. Every axis needs units — *speed (mph)*, not *speed*.

THE BLIND-SPOT PANEL — THE ONE THAT IS READ HARDEST

A blind spot is a wrong conclusion someone could reach FROM your dashboard.
Not "the data is noisy." Not "I wish I had more samples." Something a reasonable person would believe after looking at your screen, that is not true.

Good — coverage "These are point detectors on freeway mainline. This dashboard says nothing at all about ramps or arterials, and my research question is about ramp queues."	Not a blind spot "The data has some noise." <i>Every dataset has noise. What does the noise stop a viewer from concluding?</i>
Good — resolution "The data is aggregated to 5 minutes. A 90-second queue spillback is invisible here — it gets averaged away before I ever see it."	Not a blind spot "I only had time to use one year of data." <i>That is a limitation of your effort, not of what the screen can tell someone.</i>
Good — provenance "These are equilibrium flows from a model, not measurements. There is no day-to-day variation in this dataset at all, so nothing about reliability can be asked of it."	Not a blind spot "The dashboard could be improved with more charts." <i>This is a to-do list. Say what the current charts would mislead someone into believing.</i>
Good — missingness "Sensor 12 is dark for 31 hours in March and my line chart interpolates straight across it. The line looks continuous. It is not."	Why this is graded hardest. If your research question lives inside one of your own three blind spots, you do not have a project yet. Week 3 is a much cheaper place to find that out than Week 13.

BEFORE YOU SUBMIT — SELF-CHECK AGAINST THE SLIDE

- Three controls, each of which passes the numbers-change test
- Three charts, at least two kinds, units on every axis
- `@st.cache_data` on every loader, each with one sentence saying why caching it is worth it
- Controls in the sidebar; columns, tabs or an expander in the body
- Provenance **inside the app** — who collected it, where, when, with what instrument
- Blind-spot panel with three entries, each one a wrong conclusion a viewer could reach
- The URL opens in a private browser window, on a machine that is not yours

Course policy — ask me if this is not already posted:

- Which dataset: your own project data, or one of the three from class?
- Individual, or pairs?
- Submit the URL only, or the URL and the repository link?
- Where to submit, and what happens if it is late
- What you must disclose if you used AI assistance to write the code

When it breaks

CIVL 6962 · the six things that go wrong, and the one concept behind half of them

THE ONE CONCEPT: STREAMLIT RE-RUNS YOUR WHOLE SCRIPT ON EVERY INTERACTION

you move a slider

|

v

Streamlit runs app.py again, top to bottom, from scratch

|

+-> every ordinary variable is created fresh <- this is why your counter resets

+-> every @st.cache_data function is SKIPPED <- if the arguments have not changed

+-> every widget returns its CURRENT value

+-> anything in st.session_state SURVIVES <- this is the only thing that persists

|

v

the page is redrawn

So: if something must survive an interaction, it goes in `st.session_state`. If something is expensive and depends only on its arguments, it goes behind `@st.cache_data`. Everything else is recomputed every single time you touch anything, and that is normal — the cache is what makes it fast enough. `@st.cache_data` is for **data** — dataframes, arrays, anything Streamlit can copy. `@st.cache_resource` is for things you must not copy — a database connection, a loaded model. Using the wrong one usually still works, right up until it silently hands two users the same object.

THE SIX ERRORS

SYMPTOM	CAUSE	FIX
streamlit : command not found · or PowerShell refuses to run it	PATH, or the execution policy blocking the launcher script	Always start it as <code>python -m streamlit run app.py</code> . This bypasses the script entirely.
FileNotFoundError: 'data/xxx.parquet'	Relative paths resolve against the directory you launched from, not the file's own folder	<code>cd</code> into the project root first. Or make paths absolute: <code>Path(__file__).parent / "data" / "x.parquet"</code>
AttributeError: module 'streamlit' has no attribute 'cache'	<code>@st.cache</code> was removed from Streamlit	<code>@st.cache_data</code> for dataframes and arrays; <code>@st.cache_resource</code> for connections and models
deprecation warning about use_container_width	The argument was renamed	<code>width="stretch"</code>
"my variable resets every time I click something"	The rerun model above — this is not a bug	Put it in <code>st.session_state</code> , initialising it only if the key is absent
I edited the file and the browser shows the old app	Streamlit noticed but is waiting for you	Click Rerun in the top-right menu, or turn on Always rerun once and forget it

DEPLOYMENT — STREAMLIT COMMUNITY CLOUD

- `requirements.txt` must be in the **root of the repository**, not next to the app file if that file is in a subfolder. List only what your app actually imports — a package you never import can still break the build.
- Everything the app reads must be **in the repository** or downloaded at runtime. Nothing on your laptop's disk exists on the server.
- Keep data files small — well under 50 MB, and the free tier has about 1 GB of memory. Convert CSV to Parquet and downsample before you commit; a dashboard does not need every row.
- Pin nothing you do not have to, but if a build fails on a version conflict, pin the one package it names and redeploy.
- An app with no visitors goes to sleep and shows a "wake" button. **That is normal, not a failure** — it wakes when someone opens it. But check yours a day before the deadline: an app that fails to wake is an app that fails to grade.
- Test the URL in a private browser window**, signed out. An app that only works while you are logged into your own GitHub is not a public URL.

Before you ask for help, have these three things ready: the full error text (copied, not described), the exact command you ran, and the directory you ran it from. Roughly half of all Streamlit problems are answered by the last line of the traceback, and most of the rest by the second and third items.

Office hours: bring a laptop with the app already failing. Do not clean it up first — I want to see the error.